

VATSAL DHUVAD

Computer Engineering Student | Data Science | Machine Learning | Generative AI | Agentic AI
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[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Computer Engineering student specializing in Data Science, Machine Learning, AI/ML, Generative AI, Agentic AI, RAG, and Computer Vision. Skilled in Python, SQL, LangChain, LangGraph, LLM application development, local embeddings, FAISS vector search, data preprocessing, feature engineering, and model evaluation. Hands-on project experience includes Agentic RAG applications, classification, regression, clustering, and real-time OpenCV-based computer vision.

TECHNICAL SKILLS

Programming and Databases: Python, SQL, MongoDB, Firebase, Supabase

Data Science Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn

Machine Learning: Regression, Classification, Clustering, Feature Engineering, Data Preprocessing, Model Training, Model Evaluation

Algorithms and Metrics: Logistic Regression, Linear Regression, Random Forest, Decision Trees, K-Means, Accuracy, Precision, Confusion Matrix, MAE, RMSE, R-squared, Silhouette Score

Generative AI and Agentic AI: LLMs, RAG, LangChain, LangGraph, Prompt Engineering, FAISS, Hugging Face Embeddings, Groq API, Mistral Fallback

Computer Vision and Tools: OpenCV, Haar Cascade Classifiers, Streamlit, VS Code, Google Colab, GitHub

EDUCATION

B.E. Computer Engineering, LDRP-ITR, Kadi Sarva Vishwavidyalaya (KSV) - 2023 - Present
Gandhinagar, Gujarat | Currently in 4th Year (7th Semester) | CPI: 6.86

PROJECTS

Agentic RAG Career Assistant - Python, Streamlit, LangChain, LangGraph, RAG, FAISS, Hugging Face Embeddings, Groq API, Mistral Fallback

- Built an Agentic RAG-based career assistant that analyzes resumes, compares job descriptions, highlights matching skills, and supports interview preparation using LangChain and LangGraph.
- Implemented local Hugging Face embeddings with FAISS vector search, Groq LLM integration, optional Mistral fallback, Streamlit UI, PDF/DOCX parsing, and session usage limits for public deployment.

Student Placement Prediction using Machine Learning - Python, Pandas, NumPy, Scikit-learn, Matplotlib

- Built an end-to-end classification system on a 10,000-row student dataset to predict placement status using Logistic Regression.
- Performed data cleaning, label encoding, feature scaling with StandardScaler, and model evaluation using Accuracy, Precision, and a Confusion Matrix.

Real-Time Face, Eye, and Smile Detection - Python, OpenCV, Haar Cascade Classifiers, Computer Vision

- Developed a real-time computer vision application using OpenCV and webcam video capture to detect faces, eyes, and smiles from live frames.
- Implemented Haar Cascade classifiers, grayscale conversion, region-of-interest scanning, and labeled bounding boxes for detected facial features.

CERTIFICATIONS

NPTEL / IIT Madras: Python for Data Science - Elite Certification (64%)

Skill India / NSDC: Python Programming; Python for Data Science; Data Science Essentials; AI-ML Engineer

Google Cloud: Gemini for Data Scientists and Analysts; Agent Fundamentals; Engineer AI Agents with Agent Development Kit (ADK)

Anthropic Academy: Building with the Claude API; Claude Code in Action; AI Capabilities and Limitations

HP Foundation: Data Science and Analytics; AI for Beginners; Critical Thinking in the AI Era

Simplilearn: Data Structures and Algorithms in Python; AI Agent for Beginners; Intro to Generative AI Studio